

Paper III - Subspeciality of Urology (practice paper, MCh Urology Summer 2015 set)

Contents

1. III-1. Ureteric stents; ureteroscopy with or without stenting (LAQ)
2. III-2. Medical management of VUR; 8-yr boy with bilateral grade IV VUR (LAQ)
3. III-3a. Sacral neuromodulation
4. III-3b. Uses of buccal mucosa in urology
5. III-3c. HSG (USG) in neonates
6. III-3d. Penile prosthesis types and biofilms
7. III-3e. Uretero-vaginal fistula
8. III-3f. Prune belly syndrome

III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

Type: LAQ (25 marks) | ~1500 words

Introduction / History

- Ureteric stent = hollow tube placed in the ureter to keep the upper tract draining past an obstruction, injury or anastomosis (Campbell: Lower urinary tract drainage / Ureteroscopy chapters).
- **Zimskind 1967**: first indwelling silicone ureteral splint put in through a cystoscope (straight, so it tended to migrate).
- **Finney 1978: double-J** stent (a "J" at both ends). Around the same time Hepperlen described the pigtail. Self-retaining coils solved the migration problem.

Stent design

Parameter	Details
Diameter	3–8.5 Fr (adults mostly 4.7–7 Fr; 6 Fr is the usual default). Bigger stents give better extraluminal flow in extrinsic obstruction
Length	Adults 20–30 cm (usual 24–26 cm). Estimate from height (e.g. 22 cm if <5'10", 24 cm if 5'10"–6'4", 26 cm if taller) or measure on retrograde/ureteric catheter. Children: rough guide = age + 10 cm
Coils	Double-pigtail, J, multi-length/multi-coil ("one size fits a range"), loop-tail/tail stents (thin distal tail, fewer symptoms), open/closed tip
Flow	Mostly extraluminal around the stent; side holes. Note: drainage in malignant extrinsic compression is often poor → use tandem stents, metallic stents or PCN
Features	Radio-opaque, hydrophilic coating, extraction string ("stent on a string")

Diagram to draw: a double-J stent in situ (renal pelvis coil, side holes, bladder coil) with length/Fr labelled; a FECal grade I–V sketch of encrustation.

Stent materials

Material	Features
Polyethylene	Historical. Rigid and brittle, breaks; abandoned

Material	Features
Silicone	Most biocompatible, least encrustation. Soft, high friction, hard to push past tight ureters
Polyurethane	Stiff and cheap. More encrustation/irritation
Proprietary copolymers – Percuflex (Boston Scientific), C-Flex , Sof-Flex, Tecoflex	Balance strength and softness. Become softer at body temperature
Metallic – Resonance (Cook; Co-Cr-Ni-Mo alloy coil, licensed for up to 12 months)	Resists compression in malignant extrinsic obstruction. Tumour ingrowth is uncommon. Encrustation still happens
Thermo-expandable nitinol – Memokath 051	Shape-memory Ni-Ti spiral. Expands with warm saline; cooling softens it so it can be removed. For long-term stricture management
Self-expanding covered/uncovered metal stents (e.g. Allium, Uventa)	For strictures and malignant obstruction. Limited evidence
Coated	Heparin (anti-encrustation), triclosan (antimicrobial; did not convincingly reduce symptoms or infection), hydrogel (lubricious, hydrophilic), silver, phosphorylcholine, drug-eluting (e.g. ketorolac/anti-inflammatory, paclitaxel – investigational)
Biodegradable	Polymer stents that dissolve, so no removal and no "forgotten stent". Still experimental: fragmentation and uneven degradation have stopped them reaching routine use

Special stents

- **Tumour/malignant stents:** metallic (Resonance), tandem double-J, reinforced/stiff polymer.
- **Endopyelotomy stent:** tapered 14/7 Fr or 8.2/6 Fr (bulbous segment sits across the incised PUJ).
- **Magnetic-tip stent** (e.g. Black-Star/Magnetic Blackstar): removed with a magnetic catheter without cystoscopy.
- **Stent on string / extraction strings:** patient or clinic removes it. Avoids cystoscopy, but carries a risk of dislodgement.
- Others: Ureteric catheters/open-end "external" stents, multi-length stents, mono-J (urinary diversion), anti-reflux valve stents, stents with tails (Polaris Loop).

Indications

- **Relief of obstruction:** stone with sepsis (stent = PCN), malignant extrinsic obstruction, stricture, pregnancy.
- **After intervention:** URS (selected cases), endopyelotomy, pyeloplasty, ureteric reimplant/anastomosis, transplant ureteroneocystostomy.
- **Healing:** ureteric injury or leak, fistula.
- **Prophylactic/identification:** before complex pelvic surgery, before ESWL in a solitary kidney/large burden (selected), passive dilatation (pre-stenting).

Stent-related symptoms

- Up to ~80% have symptoms: frequency, urgency, suprapubic/flank pain (on voiding = reflux), haematuria, sexual dysfunction, reduced work capacity.
- **USSQ (Ureteral Stent Symptom Questionnaire; Joshi et al. 2003):** urinary symptoms, pain, general health, work performance, sexual matters, other. A validated tool.

- Mechanisms: trigonal irritation from the bladder coil crossing the midline, reflux, ureteric spasm.
- Treatment: **alpha-blockers** (tamsulosin, alfuzosin), **antimuscarinics** (solifenacin, tolterodine); the combination is better than either alone. Mirabegron, PDE5i and intravesical lidocaine have been tried. Choose correct length (bladder coil should not cross midline), soft/tail stents, shortest dwell time.

Complications

Complication	Notes
Malposition / migration	Proximal migration (too short): needs URS retrieval. Distal: symptoms or expulsion
Infection / colonisation	Colonisation is near-universal with longer dwell. Infection less common. Stent-related biofilm
Haematuria	Common and mild. Rare uretero-arterial fistula (prior radiation, vascular graft)
Encrustation	Rises with indwelling time (bigger risk >6 weeks–3 months). Risks: stone formers, UTI (urease organisms: Proteus), pregnancy, CKD, metabolic disease
Forgotten stent	Leads to encrustation, obstruction, sepsis, renal loss. Use stent registries/apps, SMS reminders
Fracture, knotting, erosion, VUR, obstruction	–

FECal (Forgotten, Encrusted, Calcified) grading (Acosta-Miranda 2009): I – minimal linear encrustation; II – circular encrustation; III – partial encrustation of one or both coils; IV – complete encrustation of one coil; V – complete encrustation of both coils. **KUB score** (Kidney-Ureter-Bladder) is an alternative.

Management of encrusted stent

- Never pull it out forcefully, because of the risk of avulsion. Assess with NCCT KUB (stone burden at each site) and DMSA if renal function is doubtful. Treat UTI first.
- Usual order: **bladder coil first** (cystolithotripsy laser/pneumatic), then **ureter** (semirigid/flexible URS laser along the shaft), then **renal coil** (fURS, or **PCNL** for large burden). ESWL is an adjunct for coil encrustation.
- Do it in a single session where possible ("endourological combined approach"). Staged procedures for heavy burden. Open/laparoscopic surgery is rarely needed. Nephrectomy if the kidney is non-functioning.
- Metabolic evaluation. Replace the stent if needed.

Ureteroscopy

Instruments

	Semirigid	Flexible (fURS)
Size	~6–8.5 Fr tip	~7.5–9.5 Fr shaft
Best for	Distal/mid ureter	Upper ureter, renal (RIRS)
Deflection	None	Active up to ~270° in both directions
Optics	Fibre-optic/rod lens	Fibre-optic → digital (chip-on-tip CMOS/CCD): better image, less fragile

- **Ureteral access sheaths (UAS)**: inner/outer 9.5/11.5, 10/12, 11/13, 12/14 Fr. Lengths ~35–55 cm.
 - Pros: lower intrarenal pressure, repeated passage, better vision, scope protection.

- Cons: **ureteral wall injury** (Traxer & Thomas 2013 grading; ~1 in 8 significant), ischaemia. Pre-stenting reduces UAS injury.
- **Lasers:**
 - **Ho:YAG (2120 nm):** gold standard. Pulsed, fragmentation + dusting, high-power with Moses (pulse-modulation) technology.
 - **Thulium fibre laser (TFL, ~1940 nm):** higher water absorption, thinner fibres (down to ~150 µm), low-energy/high-frequency settings → fine dust, less retropulsion.
- **Dusting vs fragmentation:**
 - Dusting (low energy, high frequency, no basket): faster, no UAS needed, but residual fines.
 - Fragmentation + basketing: stone-free status confirmed, material for analysis.
 - "Pop-dusting" is a variant.

Steps of URS

1. Antibiotic prophylaxis. Urine sterile.
2. Cystoscopy, safety guidewire (hydrophilic/PTFE nitinol).
3. Semirigid inspection ± dilatation (dilate only if needed).
4. UAS over working wire under fluoroscopy (for fURS).
5. Low-pressure irrigation, locate stone, laser lithotripsy, basket fragments (nitinol tipless).
6. Inspect the ureter on withdrawal (PULS: Post-Ureteroscopic Lesion Scale, Schoenthaler).
7. Decide on the stent. Postop imaging.

Stenting after URS - "with or without"

- **AUA/Endourological Society Surgical Management of Stones 2016:** clinicians **may omit** ureteral stenting after **uncomplicated** URS.
- **EAU Urolithiasis 2024:** do not routinely stent after uncomplicated URS (complete stone removal). Pre-stenting is not routinely needed but makes fURS easier and improves stone-free rate. Alpha-blockers/antimuscarinics reduce stent symptoms.
- RCTs and meta-analyses show that skipping the stent after uncomplicated URS gives fewer LUTS/pain and less cost. Unplanned visits do not rise significantly.

Indications for stenting after URS

- Ureteral injury/perforation (stent for 2–6 weeks) or significant UAS injury.
- Significant residual fragments (risk of steinstrasse).
- Solitary kidney, transplant kidney.
- Impacted stone, ureteral oedema.
- Pregnancy.
- Prolonged procedure, UTI/sepsis.
- Renal insufficiency, bilateral URS.
- Stricture/dilatation.
- Planned second-look.

Pre-stenting

- Pros: passive dilatation, so UAS passes easily, higher SFR, fewer injuries. Necessary when the ureter is too tight at first attempt.
- Cons: stent morbidity, extra procedure, cost.

Duration and removal

- Uncomplicated cases: 3–7 days, often on a **string**. String stents give less cystoscopy, similar symptoms, small risk of early dislodgement.
- Injury: 2–6 weeks. Stricture/endopyelotomy: 4–6 weeks.
- A ureteric catheter overnight is an alternative.

After URS

|

Uncomplicated? (no injury, stone-free, normal anatomy, no infection)

| -- Yes --> No stent (or stent on string 3-7 d / overnight ureteric catheter)

| -- No --> Double-J stent

- injury/perforation : 2-6 wk

- residual fragments/impacted/oedema : 1-2 wk

- solitary kidney/pregnancy/UTI/CKD : individualise

Recent advances

- **Suction ureteral access sheaths**, including **FANS (flexible and navigable suction access sheath)** with a bendable tip that follows the scope into calyces: lower intrarenal pressure and temperature, better SFR, fewer infections.
- **TFL** and pulse-modulated high-power Ho:YAG.
- **Single-use digital flexible ureteroscopes**: no reprocessing or breakage cost, sterile every time.
- Intrarenal pressure monitoring, robotic fURS platforms, AI stone recognition.
- Stents: drug-eluting and biodegradable stents (still investigational), anti-reflux stents, stent registries.

Key points to remember: - Zimskind 1967 (first indwelling stent); Finney 1978 (double-J).
- USSQ measures stent symptoms; alpha-blocker + antimuscarinic is best. FECal grades I-V for encrustation; remove bladder coil, then ureter, then kidney. - Memokath 051 = thermo-expandable nitinol; Resonance = metallic, up to 12 months. - AUA 2016 and EAU 2024: stent may be omitted after uncomplicated URS. Stent for injury, residual fragments, solitary kidney, pregnancy, impacted stone, infection. - Ho:YAG 2120 nm vs TFL ~1940 nm; UAS 10/12-12/14 Fr; FANS suction sheaths and single-use scopes are the newest additions.

III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR.

Type: LAQ (25 marks) | ~1500 words

Definition

- **Vesicoureteral reflux (VUR):** urine flows back from the bladder into the ureter/upper tract.
- **Primary:** congenital deficiency of the UVJ flap-valve. Short intravesical tunnel; the normal tunnel length : ureteric diameter ratio is ~5:1 (Paquin).
- **Secondary:** high bladder pressure or outlet obstruction (PUV, neurogenic bladder, BBD) or iatrogenic (post-incision of ureterocele).
- Prevalence ~1% of normal children, ~30–40% of children with febrile UTI, ~30% of siblings (Campbell: VUR chapter).

Pathophysiology

- Reflux plus infection allows bacteria to ascend, causing pyelonephritis. **Compound papillae** (polar) allow intrarenal reflux, which leads to **scarring**.
- Scarring causes hypertension, proteinuria and CKD (**reflux nephropathy**).
- Congenital dysplasia ("congenital reflux nephropathy") occurs especially in **boys with high-grade VUR found antenatally**.

International Reflux Study grading (VCUG; IRS 1981/1985)

Grade	VCUG findings
I	Contrast in ureter only (non-dilated)
II	Ureter, pelvis and calyces; no dilatation; normal fornices
III	Mild-moderate dilatation of ureter/pelvis; minimal blunting of fornices
IV	Moderate dilatation/tortuosity of ureter; moderate dilatation of pelvis and calyces; fornices obliterated but papillary impressions maintained
V	Gross dilatation and tortuosity; papillary impressions lost ; intraparenchymal reflux

Diagram to draw: five kidney-ureter sketches showing grades I–V. A Cohen cross-trigonal reimplant diagram.

Natural history

- Resolution depends on **grade, laterality, age at diagnosis and BBD**.
- Grades I–II: ~80% or more resolve over ~5 years.
- Grade III: ~50%. Unilateral and younger children do better.
- Grade IV: unilateral ~50–60% at 5 years; **bilateral only ~10%** (AUA 1997 meta-analysis data, Elder).
- Grade V rarely resolves.
- Annual resolution rate falls with age; most resolution happens before puberty. Nomograms (e.g. Estrada) predict this.

Evaluation

- History: febrile UTIs, **BBD** (voiding diary, constipation, Bristol stool chart, dysfunctional voiding score), family history.
- Examination: BP, growth, spine, genitalia.
- **Urine** routine/culture; serum creatinine ($\pm$ cystatin C), urine protein/creatinine ratio.

- **RBUS**: hydronephrosis, renal size, cortical thinning, bladder wall thickness, post-void residual (PVR).
- **VCUG**: grade, timing (filling/voiding), urethra (exclude PUV in boys), bladder diverticulum (Hutch), duplex. Radionuclide cystogram for follow-up (lower radiation). Contrast-enhanced voiding urosonography (ceVUS) is an option.
- **DMSA**: scars and differential renal function (DRF); acute DMSA confirms pyelonephritis.
- **Uroflowmetry + PVR ± urodynamics** if BBD or secondary VUR suspected.

Medical management

1. Continuous antibiotic prophylaxis (CAP) (single nightly dose, ~¼ therapeutic dose)

Drug	Dose	Notes
Trimethoprim-sulfamethoxazole	2 mg/kg TMP (with 10 mg/kg SMX) once daily	Avoid <2 months (kernicterus)
Trimethoprim alone	2 mg/kg	-
Nitrofurantoin	1-2 mg/kg nightly	GI upset; avoid in CKD/G6PD deficiency
Cephalexin / amoxicillin	Cephalexin ~10 mg/kg	Neonates/infants <2 months

2. Evidence for CAP

- **RIVUR (NEJM 2014)**: 607 children aged 2–71 months with VUR grade I–IV after febrile/symptomatic UTI; TMP-SMX vs placebo.
 - CAP **halved recurrent UTI** (HR ~0.50); greatest benefit with **BBD** and febrile index UTI.
 - **No difference in new renal scarring**. More resistant organisms in breakthrough infections.
- **Swedish Reflux Trial (Brandström 2010)**: children 1–2 years with grade III–IV VUR; CAP vs endoscopic treatment vs surveillance.
 - In **girls**, CAP and endoscopy reduced recurrent febrile UTI, and new scars were more frequent on surveillance.
 - **Boys** had few recurrences whatever the arm.
- **PRIVENT (Craig, NEJM 2009)**: TMP-SMX in children with prior UTI (~40% had VUR) gave a modest absolute reduction (~6%) in symptomatic UTI.

3. Bladder and bowel dysfunction (BBD) treatment

- Treat this first in toilet-trained children.
- Timed voiding (every 2–3 h), double voiding, hydration, posture.
- Treat constipation (PEG laxatives). Urotherapy/biofeedback.
- Anticholinergics for overactivity; alpha-blockers for dysfunctional voiding. CIC if needed.
- BBD lowers resolution rates, increases breakthrough UTI and reduces endoscopic success.

4. Circumcision: reduces UTI risk in **infant boys**. Recommended by EAU/ESPU for boys with VUR in the first year.

5. Surveillance without CAP: an option for low-grade VUR without BBD in older children, with prompt treatment of febrile UTI.

6. Follow-up

- Urine testing for any fever.
- RBUS every 6–12 months.
- VCUG/radionuclide cystogram every 12–24 months.
- DMSA if breakthrough UTI.
- BP, growth, proteinuria (lifelong if scarred).

Guidelines

- **AUA VUR guideline 2010 (children >1 year):**
 - CAP recommended for child with **BBD + VUR**; may be considered without BBD.
 - Treat BBD first. Recommends DMSA, BP, height, weight, urinalysis.
 - Breakthrough febrile UTI on CAP: change therapy, consider intervention.
- **EAU/ESPU (Paediatric Urology guideline):** risk stratified.
 - **High-risk** (grade IV–V, scars, BBD, febrile UTI, abnormal kidneys): CAP initially; **early intervention** if breakthrough UTI or persistent high grade.
 - Low-risk: surveillance or CAP.
 - Intervention prefers open/laparoscopic reimplant for high grade and endoscopic injection for low-moderate.

Indications for intervention

- Breakthrough febrile UTI on CAP.
- New or progressive scarring / deteriorating DRF.
- Persistent high-grade (IV–V) VUR unlikely to resolve, especially in an older child.
- Non-compliance or intolerance of CAP.
- Associated anomalies (paraureteric diverticulum, duplex).
- **Parental preference.**
- Persistent VUR at puberty in girls (pregnancy risks).

Case: 8-year-old boy with bilateral grade IV VUR

Assessment

- Is this primary or secondary? Exclude **PUV** (VCUG urethra), neurogenic bladder (spine examination, lumbosacral X-ray/MRI if signs) and **BBD** (diary, uroflow, PVR).
- Febrile UTI history and breakthrough infections.
- **DMSA** for scars and DRF. Creatinine/eGFR, **BP**, proteinuria.
- Bilateral grade IV at age 8 → **spontaneous resolution chance is low (~10%)**, and few years remain before puberty.

Options

1. **Continued CAP + BBD management.**
 - Reasonable if he has no febrile UTIs, no scars and good bladder function. Boys, especially circumcised boys past infancy, have a lower UTI risk.
 - Downsides: long-term CAP, resistance, compliance, low resolution rate.
2. **Endoscopic injection (Deflux - dextranomer/hyaluronic acid copolymer; FDA approved 2001).**
 - Techniques: STING (subureteric) or **HIT/double-HIT** (hydrodistension-implantation, intraureteric).
 - **Elder meta-analysis (J Urol 2006):** success after one injection by grade: I–II ~79%, III ~72%, **IV ~63%**, V ~51%. Overall success rises with repeat injection.
 - Day-care procedure, minimal morbidity. Risks: lower durability, late recurrence, rare late obstruction, calcified mounds that mimic stones.
 - Lower success in BBD and duplex systems.
3. **Ureteral reimplantation** (gold standard for high grade; success ~**95–98%**, slightly lower in grade V).
 - **Intravesical:**
 - **Cohen cross-trigonal:** most popular, simplest, very reliable; later ureteric access for URS is difficult.
 - **Politano-Leadbetter:** new suprahial hiatus; risk of bowel injury and ureteric kinking.

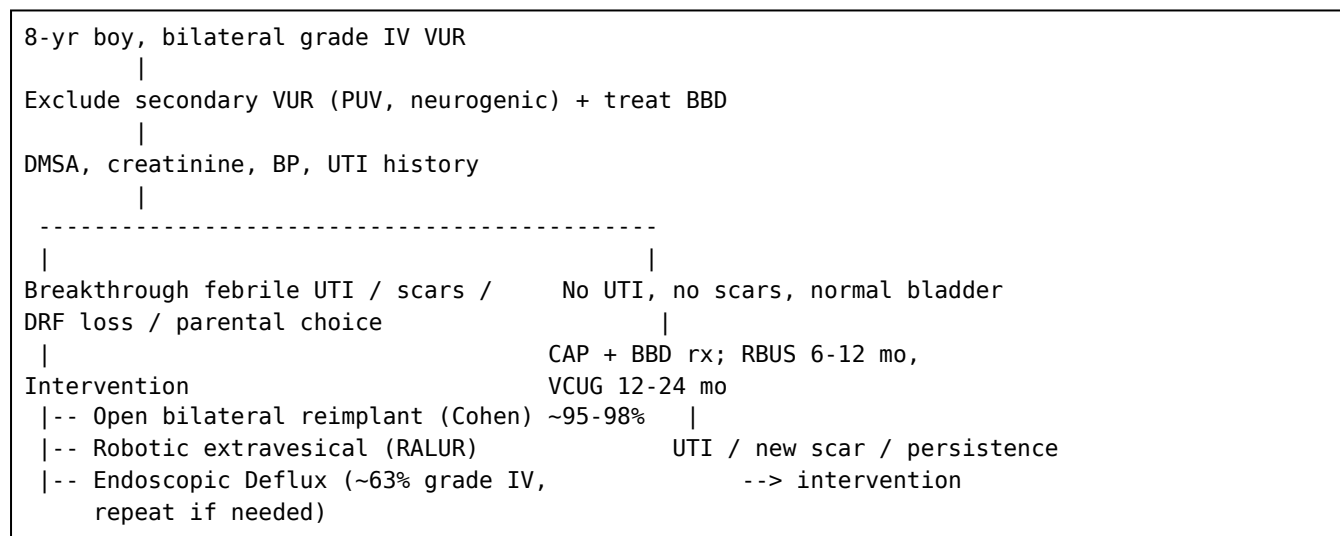
- **Glenn-Anderson**: advancement towards the bladder neck.
- **Extravesical (Lich-Gregoir; detrusorrhaphy)**: no ureterotomy, less haematuria/spasm, shorter stay. **Bilateral extravesical carries a risk of transient urinary retention** (pelvic plexus injury), reduced by nerve-sparing dissection.
- **Robotic-assisted laparoscopic ureteral reimplantation (RALUR, extravesical)**: less pain, shorter stay. Success in series is ~85–95%, somewhat lower and more variable than open. Bilateral retention risk similar.
- Bilateral intravesical **Cohen** is ideal for bilateral grade IV because one bladder opening corrects both sides. Or extravesical/robotic in experienced hands.

Recommended plan

- Explain options to parents (shared decision).
- If **febrile UTIs/breakthrough, scars, or DRF loss** → **bilateral reimplantation** (open Cohen, or robotic extravesical by expertise). Endoscopic injection is an acceptable lower-morbidity alternative, accepting ~60–70% success per session.
- If **asymptomatic, no scars, normal bladder** → CAP + BBD therapy with yearly imaging; intervene if UTI or new scar.
- Treat BBD before any surgery.

Postoperative care and follow-up

- Catheter 1–2 days (intravesical: ± stents/suprapubic catheter). Anticholinergics for spasms.
- Continue CAP ~3 months.
- RBUS at 4–6 weeks and ~1 year for obstruction/hydronephrosis.
- Postop VCUG is optional after open surgery (AUA: may be omitted if RBUS normal); usually done after endoscopic injection.
- Long-term: BP, urine protein, renal growth; DMSA if scarred.
- Complications: persistent/contralateral VUR (new contralateral VUR after unilateral reimplant), obstruction, bleeding.



Recent advances

- **ceVUS** (contrast-enhanced voiding urosonography) avoids radiation.
- "Top-down" approach (DMSA first) in febrile UTI.
- Polyacrylate-polyalcohol copolymer (Vantris) as an alternative bulking agent.
- RALUR outcomes improving.
- Selective rather than universal CAP after RIVUR; urinary biomarkers for scarring.

Key points to remember: - IRS grading I-V: grade IV keeps papillary impressions, grade V loses them. - RIVUR 2014: CAP halves recurrent UTI but does not change scarring; benefit greatest with BBD. - CAP doses: TMP 2 mg/kg; nitrofurantoin 1-2 mg/kg nightly. - Deflux success per injection (Elder 2006): grade IV ~63%. Open reimplant ~95-98%. - Bilateral extravesical reimplant carries a transient retention risk; Cohen is the intravesical workhorse.

III-3a. Sacral neuromodulation

Type: SAQ (10 marks) | ~550 words

Definition and mechanism

- **Sacral neuromodulation (SNM)**: continuous low-amplitude electrical stimulation of the **S3 nerve root** through a tined lead and an implantable pulse generator (IPG) (Campbell: Neuromodulation chapter).
- Mechanism: **afferent modulation**, not direct motor stimulation.
 - Stimulating somatic afferents inhibits abnormal bladder afferent reflexes at spinal and supraspinal levels (pontine micturition centre, cortical changes on fMRI). This explains its effect on **both overactivity and retention**.
 - In retention it is thought to suppress the inappropriate "guarding reflex" of the urethral sphincter.

Indications and FDA approvals

Indication	FDA
Refractory urgency urinary incontinence	1997
Urgency-frequency	1999
Non-obstructive urinary retention (incl. Fowler's syndrome)	1999
Faecal incontinence	2011

- Off-label: IC/BPS, selected neurogenic LUTD.
- **Fowler's syndrome**: young women with retention, high residuals and abnormal sphincter EMG (complex repetitive discharges/decelerating bursts). SNM is the best treatment.

Devices

- **Medtronic InterStim**: InterStim II (non-rechargeable), **InterStim Micro** (rechargeable, small), **InterStim X** (recharge-free, longer battery).
- **Axonics** (rechargeable r-SNM, and recharge-free F15; now part of Boston Scientific).
- Newer systems are **full-body MRI conditional** (1.5 T and 3 T, check labelling).

Technique

1. Test phase

- **PNE (percutaneous nerve evaluation)**: temporary unipolar wire, 3–7 days, office procedure. Higher false-negative rate from migration.
- **Staged (stage 1) tined lead**: permanent quadripolar tined lead under fluoroscopy, connected to an external stimulator, trial ~1–2 weeks. More reliable.

2. Prone position. Needle into **S3 foramen** (≈ 9 cm above the coccyx tip, ~2 cm lateral to the midline; fluoroscopic landmarks).

3. Responses

- **S3: bellows** (pelvic floor contraction/deepening of the gluteal cleft) + **big toe plantar flexion**. Sensory: pulling in rectum/vagina/perineum.
- S2: "clamp" of anal sphincter, whole-foot plantar flexion/leg rotation.
- S4: bellows only, no toe response.

4. Aim for motor response on ≥3 of 4 electrodes at low amplitude (curved stylet).

5. **Success criterion**: **≥50% improvement** in key bladder-diary symptoms (leaks, voids, catheterised volume), then stage 2 IPG implant in the upper buttock.

Outcomes

- **InSite trial** (Siegel 2015): SNM gave higher therapeutic success than standard medical therapy at 6 months in mild-moderate OAB, with durable benefit on 5-year follow-up.
- **ROSETTA trial** (Amundsen, JAMA 2016): onabotulinumtoxinA **200 U** vs SNM in women with refractory UUI.
 - Botox gave a **slightly greater reduction in daily UUI episodes at 6 months** (statistically significant, small difference) and higher satisfaction.
 - Botox caused **more UTIs** and need for **CISC** (transient retention).
 - SNM group had device revisions/removals.
- Retention: ~70% of responders void spontaneously.

Complications

- Pain at IPG site.
- Lead migration, loss of efficacy, infection (requires explant), unwanted stimulation/shock, battery depletion.
- **Revision rates** historically ~30% or higher with older techniques; lower with the tined lead and newer systems.

Contraindications

- Failed test phase.
- Inability to use the programmer or keep a diary, cognitive impairment.
- Anatomical sacral abnormality (e.g. sacral agenesis) or spinal pathology that prevents lead placement.
- Need for repeated MRI with older non-conditional devices.
- Relative: pregnancy (switch off), bleeding disorder.

Recent advances

- Rechargeable and recharge-free miniaturised IPGs with full-body MRI compatibility.
- Closed-loop/ECAP-sensing concepts (investigational).
- **Tibial nerve stimulation** as an alternative: PTNS, and implantable tibial devices (eCoin, Revi).

Diagram to draw: sacrum with the tined lead in the S3 foramen and the IPG in the buttock; S3 response table.

Key points to remember: - FDA approvals: 1997 UUI, 1999 urgency-frequency and non-obstructive retention, 2011 faecal incontinence. - Mechanism is afferent modulation. S3 response = bellows + big-toe plantar flexion. - Success in the test phase means $\geq 50\%$ improvement; staged tined lead beats PNE. - ROSETTA 2016: Botox 200 U slightly better than SNM for UUI at 6 months, with more UTIs/CISC. - Fowler's syndrome responds best to SNM.

III-3b. Uses of buccal mucosa in urology

Type: SAQ (10 marks) | ~550 words

Introduction / history

- **Buccal mucosal graft (BMG)** is the workhorse free graft for urethral reconstruction (Campbell: Urethral stricture/Reconstruction chapter).
- **Humby 1941**: first used for hypospadias.
- **Bürger 1992** and **El-Kasaby 1993** popularised it for urethral stricture.
- **Morey & McAninch 1996**: ventral onlay bulbar.
- **Barbagli 1996**: dorsal onlay.

Advantages

- Thick, **non-keratinised** stratified squamous epithelium. Tough and resists infection.
- **Thin, highly vascular lamina propria** with a **panlamina vascular plexus**: rapid imbibition and inosculation, good take.
- **Hairless**, used to a **wet** environment, easy to harvest, concealed donor scar.
- Compatible with lichen sclerosus (unlike genital skin).

Harvest technique

- Nasotracheal intubation preferred. Retractor/bite block.
- Mark the inner cheek graft (~4–6 cm × 2–2.5 cm per cheek; bilateral if needed). Stay **1 cm from the lip commissure** and **avoid Stensen's (parotid) duct opening opposite the upper 2nd molar**.
- Infiltrate with saline ± adrenaline, excise mucosa without muscle, achieve haemostasis. Close the donor site or leave it open (both acceptable; evidence mixed).
- **Defat** the graft on the back table.
- **Alternative: lingual mucosal graft** (ventrolateral tongue; Simonato 2006). Others: lip (avoid, perioral numbness).

Donor-site complications

- Pain, swelling, perioral numbness.
- Difficulty mouth opening (contracture, trismus).
- Salivary changes, Stensen's duct injury, bleeding.
- Mostly transient.

Uses

Area	Use
Anterior urethroplasty (main use)	Bulbar : dorsal onlay (Barbagli), ventral onlay (Morey/McAninch), dorsal inlay (Asopa 2001) via ventral sagittal urethrotomy, Kulkarni one-sided dorsal onlay (2009; preserves ventral blood supply and the bulbospongiosus), augmented anastomotic urethroplasty (excise the obliterative segment + graft the rest). Penile : dorsal inlay (Asopa) or dorsal onlay
Staged urethroplasty	Johanson (1953) and Bracka (1995) principle: stage 1 = lay open + graft; stage 2 = tubularise after ~6 months. For lichen sclerosus , failed hypospadias, long panurethral/infected strictures
Hypospadias	Redo/"hypospadias cripples" (staged Bracka), lack of local skin

Area	Use
Bladder neck contracture / posterior urethral (vesico-urethral anastomotic) stenosis	Robotic BMG onlay reconstruction , an emerging technique
Ureter	Robotic BMG ureteroplasty (Zhao et al. ; onlay with omental/perirenal fat wrap) for long proximal/mid strictures where excision + anastomosis is not possible. Avoids ileal ureter/autotransplant
Others	Vaginoplasty (MRKH, DSD, gender-affirming), urethral reconstruction in women (dorsal/ventral onlay for female stricture), Peyronie's grafting (rarely), Mitrofanoff/neourethra in exstrophy (selected)

Outcomes

- Anterior BMG urethroplasty success **~85-90%** (dorsal vs ventral onlay similar in the bulbar urethra).
- Results are lower in penile, lichen sclerosus and redo cases.
- Robotic BMG ureteroplasty: short-term success ~90% in early series.

Recent advances

- Tissue-engineered oral mucosa (MukoCell) as an investigational alternative.
- Robotic BMG for ureter and bladder neck.
- Non-transecting bulbar techniques combined with grafts.
- Single-cheek harvest with no closure.

Diagram to draw: cross-section of the bulbar urethra showing dorsal onlay (graft on corpora), ventral onlay and Asopa inlay; inner cheek showing Stensen's duct and harvest site.

Key points to remember: - Humby 1941 (hypospadias); Bürger 1992 / El-Kasaby 1993 (stricture). - Thin lamina propria + panlamina plexus = excellent graft take. - Avoid Stensen's duct opposite the upper 2nd molar; lingual graft is the alternative. - Barbagli dorsal onlay, Asopa dorsal inlay, Kulkarni one-sided dorsal; staged Johanson/Bracka for lichen sclerosus. - Robotic BMG ureteroplasty (Zhao) handles long ureteric strictures. Anterior urethroplasty success ~85-90%.

III-3c. HSG in neonates

Type: SAQ (10 marks) | ~550 words

Assumption: hysterosalpingography is not performed in neonates, so this is taken as a likely typo for "USG (ultrasonography) in neonates".

Introduction

- **First-line, radiation-free** neonatal urinary imaging; main use is **antenatal hydronephrosis (ANH)**, seen in ~1-2% of pregnancies (Campbell: Perinatal urology chapter).

Indications and timing

- Indications: ANH or other antenatal anomaly; febrile UTI; abdominal mass; oligohydramnios, single umbilical artery or syndromes (VACTERL, anorectal malformation, spina bifida); poor stream/distended bladder; hypertension, renal failure.
- **Avoid the first 48-72 h:** physiological oliguria **underestimates** dilatation.
- **Exception:** scan **within 24-48 h** for suspected **PUV**, bilateral severe HDN, solitary kidney with HDN, or oligohydramnios.
- If the first scan is normal or mild, **repeat at 4-6 weeks**.
- Technique: 5-10 MHz probe, longitudinal + transverse views, **APD on transverse view at the hilum**, calyces, parenchyma, ureters, full bladder (wall, ureterocele), PVR, Doppler.

Normal neonatal kidney

- Length ~4-5 cm.
- **Prominent hypoechoic medullary pyramids** (do not mistake for hydronephrosis).
- **Cortex isoechoic or slightly hyperechoic to liver** (an adult kidney is hypoechoic).
- Fetal lobulation. Little sinus fat. Bladder: wall thickness, diverticulum, megacystis.

Grading of dilatation

- **APD:** postnatal ≥ 10 mm is abnormal; ≥ 15 mm moderate; > 20 mm severe (rough guide).
- **SFU (Society for Fetal Urology) grading 0-4:** 0 none; 1 pelvis only; 2 pelvis + some calyces; 3 all calyces; 4 grade 3 + parenchymal thinning.

UTD classification 2014 (multidisciplinary consensus, Nguyen et al.) - postnatal (≥ 48 h)

Grade	Findings
Normal	APD < 10 mm, no calyceal dilatation
UTD P1 (low risk)	APD 10-< 15 mm or central calyceal dilatation; normal parenchyma, ureters and bladder
UTD P2 (intermediate)	APD ≥ 15 mm , peripheral calyceal dilatation, or ureteral dilatation
UTD P3 (high risk)	P2 features + abnormal parenchyma (thickness, echogenicity, cysts) or abnormal bladder

- Antenatal: **A1** (16-27 wk APD 4- < 7 mm; ≥ 28 wk 7- < 10 mm) and **A2-3** (≥ 7 / ≥ 10 mm or any other abnormality).

Findings in specific conditions

Condition	USG findings
PUJO	Pelviccalyceal dilatation with normal ureter and bladder . Pelvis disproportionate to calyces
VUR	Fluctuating dilatation, urothelial thickening; normal USG does not exclude VUR
PUV	Bilateral hydroureteronephrosis, thick-walled bladder , dilated posterior urethra (transperineal), echogenic/cystic kidneys, urinoma. " Keyhole sign " is an antenatal USG sign; postnatal confirmation is by VCUG
MCDK	Non-communicating cysts, no normal parenchyma; check contralateral kidney
Duplex / ureterocele	Two sinus echo complexes, upper-pole hydro; thin-walled intravesical cyst
ARPKD	Bilaterally enlarged, diffusely echogenic kidneys , loss of corticomedullary differentiation (CMD), tiny cysts; hepatic fibrosis
Adrenal haemorrhage	Suprarenal mass, avascular on Doppler , evolves on serial scans (neuroblastoma is vascular)
Renal vein thrombosis	Enlarged echogenic kidney, loss of CMD, absent venous flow; triad: haematuria, flank mass, thrombocytopenia

Follow-up algorithm

Antenatal HDN Postnatal USG (day 3-7; within 24-48 h if PUV/bilateral/solitary/oligohydramnios) Normal --> repeat USG 4-6 wk --> normal: discharge P1 --> USG 1-6 mo P2 --> USG 1-3 mo, +/- VCUG, MAG3 if persists, consider CAP P3 --> USG 1 mo, VCUG, MAG3/DMSA, CAP; urgent if PUV (catheter, valve ablation)

If the examiner meant genitography/contrast studies in DSD neonates: retrograde contrast via the perineal opening defines the urogenital sinus and vaginal confluence level. It complements pelvic USG and endoscopy.

Recent advances

- ceVUS instead of VCUG; MR urography for complex anatomy; AI-assisted HDN grading.

Diagram to draw: the SFU grade 0-4 kidney sketches and the transverse APD measurement.

Key points to remember: - Avoid scanning in the first 48-72 h (oliguria) except for suspected PUV/bilateral disease; repeat at 4-6 weeks. - Neonatal kidney: prominent hypoechoic pyramids and relatively echogenic cortex are normal. - UTD 2014: P1 APD 10-15 mm; P2 ≥15 mm, peripheral calyces or ureter; P3 adds parenchymal or bladder abnormality. - Keyhole sign = antenatal USG (or VCUG) sign of PUV.

III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis?

Type: SAQ (10 marks) | ~550 words

Introduction

- Penile prosthesis = third-line treatment for erectile dysfunction refractory to or unsuitable for PDE5i/intracavernosal therapy (AUA ED guideline 2018; Campbell: Surgery for ED chapter).
- Satisfaction is the highest of all ED therapies: **>85-90%** patient and partner.

Types

Type	Examples	Pros	Cons
Semirigid / malleable	Coloplast Genesis , Boston Scientific Spectra , Tactra ; Rigicon Rigi10	Simple, cheap (useful in India), no mechanical failure, good for poor dexterity, holding condom catheter	Constantly rigid, concealment hard, erosion risk, less natural
2-piece inflatable	AMS Ambicor (pump-reservoir in scrotum)	No abdominal reservoir (prior pelvic surgery, transplant)	Less flaccidity and girth
3-piece inflatable (IPP)	AMS 700 CX (girth expansion), LGX (length + girth), CXR (narrow, for fibrosis/small corpora); Coloplast Titan , Titan Touch , Titan narrow base ; newer: Rigicon Infla10, Zephyr (outside USA)	Best rigidity and flaccidity, most natural	Cost, mechanical failure, reservoir complications

Components of 3-piece IPP

- Paired cylinders in the corpora (measured, plus rear-tip extenders).
- Scrotal pump (with lock-out valve to prevent autoinflation).
- Reservoir: **space of Retzius** (traditional) or **ectopic/submuscular**, i.e. high submuscular under the transversalis fascia/rectus (after cystectomy/prostatectomy/transplant).
- Approaches: penoscrotal (most common) or infrapubic.
- Mechanical survival ~ **80-90% at 5 years** and 60-70% at 10 years (approx.).

Coatings

- **AMS InhibiZone**: rifampin + minocycline impregnated.
- **Coloplast**: **hydrophilic coating** that absorbs the antibiotic solution the surgeon dips it in (e.g. rifampin/gentamicin, vancomycin/gentamicin).
- Both reduced infection rates to roughly half compared with uncoated devices.

Biofilm

Definition

- A **structured community of microorganisms enclosed in a self-produced extracellular polymeric substance (EPS) matrix**, adherent to a surface (the implant).
- Organisms are phenotypically altered (slow-growing, persister cells).

Stages

1. **Conditioning film**: host proteins (fibrinogen, fibronectin) coat the device within minutes.
2. **Reversible then irreversible adhesion** (adhesins, polysaccharide intercellular adhesin).

3. **Microcolony formation.**
4. **Maturation:** 3D mushroom architecture with water channels and quorum sensing.
5. **Dispersal/detachment:** planktonic seeding causes clinical infection.

Organisms

- **Coagulase-negative staphylococci (S. epidermidis)** dominate overall, often with indolent infection.
- **S. aureus, Gram-negatives** (E. coli, Pseudomonas, Enterobacter).
- Increasingly virulent and **anaerobes, Candida** (diabetics).
- Polymicrobial infections are common.

Why antibiotics fail

- EPS limits penetration. Metabolically dormant persister cells. Altered microenvironment (pH, O₂).
- MIC may be 100–1000× higher than for planktonic cells.
- Biofilm is present on many clinically uninfected devices too (subclinical).

Infection rates

- **~1-3% virgin implants** (lower, <1-2%, with coated devices).
- Higher in **revision surgery, diabetics with poor control**, spinal cord injury, immunosuppression, prolonged surgery and corporal fibrosis.

Prevention

- Treat UTI and skin issues preoperatively; good glycaemic control (HbA1c threshold debated).
- Chlorhexidine-alcohol skin prep; IV antibiotics (vancomycin/cefazolin + aminoglycoside).
- **"No-touch" technique** (Eid): isolate skin with drapes, change gloves.
- **Antibiotic-coated device**, antibiotic irrigation/washout, short operating time, minimal traffic.
- Avoid haematoma (closed suction drain debated).

Management of infection

- **Mulcahy salvage protocol** (1996): remove all components and foreign material, then sequential **7-solution antiseptic washout** (antibiotic, hydrogen peroxide, povidone-iodine, etc.), then immediate replacement. Success **~80-90%**.
- Not suitable with sepsis, necrosis, erosion, diabetic ketoacidosis or immunocompromise.
- **Explant** + delayed reimplant (fibrosis makes it difficult) or immediate malleable "placeholder" to preserve length.

Recent advances

- Tactra malleable (2019) and Rigicon Infla10 IPP.
- Antibiotic-dip protocols.
- Ectopic reservoir placement as standard after pelvic surgery.

Diagram to draw: 3-piece IPP in situ (cylinders, scrotal pump, Retzius reservoir); biofilm life cycle (5 stages).

Key points to remember: - Types: malleable (Genesis, Tactra/Spectra), 2-piece (Ambicor), 3-piece (AMS 700 CX/LGX/CXR, Titan/Titan Touch). - InhibiZone = rifampin + minocycline; Coloplast = hydrophilic coating dipped in antibiotic. - Biofilm = microbial community in an EPS matrix; S. epidermidis is commonest; MIC up to 1000× higher. - Infection ~1-3% virgin, higher in revision/diabetic patients. - Mulcahy salvage (washout + immediate replacement) succeeds in ~80-90%.

III-3e. Uretero-vaginal fistula

Type: SAQ (10 marks) | ~500 words

Definition

- Abnormal communication between the ureter and the vagina, causing **continuous urinary leakage per vaginum** (Campbell: Urinary tract fistulae chapter).

Aetiology

- Gynaecological surgery, esp. hysterectomy** (abdominal > laparoscopic/vaginal): **most common in developed countries**.
 - Injury at the uterine artery crossing, cardinal ligament or vaginal cuff (clamping, ligation, diathermy/energy device thermal injury, devascularisation).
- Obstetric** (caesarean, caesarean hysterectomy, obstructed labour, instrumental delivery): **common in developing countries** including India.
- Others: pelvic malignancy (cervical carcinoma), **radiation**, colorectal/vascular surgery, trauma, ureteroscopy.

Presentation

- Continuous leak with normal voiding** (the bladder fills from the contralateral kidney), typically **1-4 weeks after surgery** (thermal/ischaemic necrosis).
- Flank pain, fever, ileus, urinoma, vaginal discharge.
- Obstruction/hydronephrosis in many cases.

Investigations

- Three-swab / tampon test with dye**.
 - Methylene blue instilled into the bladder: **blue upper swab = VVF**.
 - Wet but unstained (clear) swab = UVF**.
 - Oral **phenazopyridine** (orange urine) confirms a ureteric source.
- CT urography** (with delayed phase): contrast extravasation into the vagina, hydroureteronephrosis, urinoma.
- Cystoscopy + retrograde pyelogram**: site and length of injury, whether a guidewire can pass. Retrograde stent attempt in the same sitting.
- Antegrade nephrostogram via PCN if retrograde access fails.
- Exclude coexisting VVF (~10-15%)** and bilateral injury.
- Serum creatinine, urine culture. Biopsy if malignancy is suspected.

Management

Principles: drain the kidney, preserve renal function, treat sepsis/urinoma, exclude malignancy.

1. Endourological (first line if ureteric continuity is preserved)

- Retrograde double-J stent** (success when a guidewire crosses the defect).
- Failed retrograde: **PCN + antegrade stent**, or **rendezvous** (antegrade + retrograde).
- Stent for ~6-8 weeks, then RGP/CT. Many fistulae heal. Stricture may follow and need dilatation.
- PCN alone when stent placement fails, as a temporising measure.

2. Surgical repair (stent failure, complete transection, ligation, stricture)

- Injury is usually **distal (juxtavesical)**, so **ureteroneocystostomy** (refluxing or non-refluxing).

- **Psoas hitch:** bridges ~5 cm (up to ~6–10 cm in some series).
- **Boari flap:** bridges ~10–15 cm.
- Mid-ureter: ureteroureterostomy (rarely suitable here).
- Long defects: transureteroureterostomy, ileal ureter, autotransplant.
- **Timing: early repair (within the first few weeks, if diagnosed early and the patient is stable with no sepsis)** is now preferred over the traditional 3–6 month wait. Outcomes are similar and suffering is less. Delay if there is infection, radiation injury or a poor general condition.
- **Approach:** open, **laparoscopic or robotic-assisted** reimplant with psoas hitch/Boari (less morbidity, comparable success).
- Radiation fistula: wider excision, omental interposition, sometimes diversion.

```

Post-hysterectomy continuous leak + normal voiding
|
3-swab test -> clear wet swab => UVF (blue => VVF)
|
CTU + cystoscopy/RGP (exclude VVF, bilateral)
|
Guidewire passes? --Yes--> DJ stent 6-8 wk --> healed / stricture -> dilate/reimplant
|No
PCN +/- antegrade/rendezvous stent
|fails
Ureteroneocystostomy + psoas hitch / Boari (open/lap/robotic), early if fit

```

Prevention

- Identify the ureter during hysterectomy. Use selective prophylactic stents/illuminated stents in complex cases.
- Intraoperative cystoscopy with dye; ICG fluorescence.

Recent advances

- Robotic reimplant.
- Intra-ureteric ICG/near-infrared identification.
- Metallic/long-term stents in malignant/radiation cases.

Diagram to draw: the three-swab test; psoas hitch ureteroneocystostomy.

Key points to remember: - Hysterectomy is the commonest cause; obstetric causes predominate in developing countries. - Continuous leak + normal voiding; the clear wet swab on three-swab test means UVF. - Exclude coexisting VVF (~10–15%). - Stent first if the wire passes; otherwise reimplant with psoas hitch/Boari, early if fit.

III-3f. Prune belly syndrome

Type: SAQ (10 marks) | ~550 words

Definition

- **Eagle-Barrett syndrome / triad syndrome** (Campbell: Prune belly syndrome chapter).
Triad:
 1. **Deficient abdominal wall musculature** (wrinkled "prune" abdomen).
 2. **Urinary tract dilatation** (ureters, bladder, prostatic urethra).
 3. **Bilateral intra-abdominal undescended testes**.
- Incidence ~**1 in 30,000-50,000 live births**; ~**95% male**.

Pathogenesis (theories)

- **Mesodermal arrest** (~6-10 weeks): abnormal lateral plate/intermediate mesoderm affects abdominal wall muscle, urinary tract smooth muscle and gubernaculum.
- **Urethral obstruction theory**: transient fetal outlet obstruction (prostatic hypoplasia, urethral atresia/valve-like obstruction) causes massive bladder distension, which compresses the abdominal wall and blocks testicular descent.
- Yolk-sac theory (less accepted).
- **Genetics**: mostly sporadic. **CHRM3** (muscarinic receptor) mutations in familial cases; HNF1B, FLNA (X-linked) reported. Increased in twins.

Genitourinary features

- **Kidneys**: variable dysplasia (determines prognosis), hydronephrosis.
- **Ureters**: massively **dilated, tortuous**, poor peristalsis (especially distally), with VUR in ~75%.
- **Bladder**: **large, smooth-walled**, high capacity, poor contractility, large PVR. **Urachal remnant/pseudodiverticulum** at the dome.
- **Prostatic hypoplasia**: dilated prostatic urethra, triangular on VCUG. ± anterior urethral anomalies (megalourethra, atresia).
- **Testes**: bilateral **intra-abdominal** (near the iliac vessels/over the ureters). Infertility (nearly all; ART reported). Malignancy risk.

Extra-genitourinary features

- **Pulmonary**: hypoplasia (oligohydramnios), pneumothorax, ineffective cough, recurrent infections.
- **Cardiac**: ~10% (PDA, VSD, ASD, tetralogy).
- **Orthopaedic**: ~45% (talipes, hip dislocation, scoliosis, pectus).
- **GI**: malrotation, atresia, Hirschsprung, constipation.

Woodard classification

Category	Features	Outlook
I	Oligohydramnios, pulmonary hypoplasia , renal dysplasia, Potter facies, ± urethral atresia	Stillbirth/neonatal death
II	Full triad, moderate-severe uropathy, no pulmonary hypoplasia , renal function variable/impaired	Risk of urosepsis and CKD; may need diversion/reconstruction
III	Incomplete/mild triad, mild uropathy, stable renal function	Good; conservative care

Management

Neonatal

- Respiratory support. Assess renal function (creatinine after day 2–3; nadir creatinine is prognostic).
- USG, **VCUG with caution** (sepsis risk; strict asepsis + antibiotics), DMSA/MAG3. CAP.
- Avoid unnecessary instrumentation.

Urinary drainage

- Most category II–III children are managed **conservatively**: CAP, timed/double voiding, CIC if large PVR.
- Indications for drainage: infection, rising creatinine, obstruction.
 - **Cutaneous vesicostomy** (Blocksom) is preferred.
 - Cutaneous pyelostomy/ureterostomy if upper tract obstruction.
 - Internal urethrotomy (Cukier) for functional outlet obstruction (selected).

Reconstruction (selected cases)

- **Reduction cystoplasty** (excise the urachal dome; limited long-term benefit).
- **Ureteral tailoring** + reimplantation (excise the poor distal ureter) for VUR/obstruction.
- Mitrofanoff for CIC.

Orchiopexy

- **Transabdominal** approach (open or laparoscopic), **preferably by 6–12 months** (early, while the testicular vessels are mobile).
- **Fowler-Stephens** (single or two-stage laparoscopic) for short vessels.
- Combine with abdominoplasty/reconstruction in one sitting.

Abdominoplasty

- Cosmetic and functional (better cough, voiding, bowel function).
- **Randolph** (transverse; 1981), **Ehrlich** ("vest-over-pants"; 1986), **Monfort** (umbilicus-sparing central plication with fascial overlap; 1991, the most popular).

Prognosis

- About one-third develop **CKD** in childhood/adolescence.
- **Renal transplantation** gives results similar to other causes (assess bladder, CIC if needed).

Females and incomplete forms

- Females (~5%): abdominal wall defect + urinary anomalies (no testicular component).
- **"Pseudo-prune belly"**: incomplete triad (normal abdominal wall with urinary/testicular features, or vice versa); usually milder.

Recent advances

- Fetal vesicoamniotic shunting in selected cases (benefit unproven).
- Laparoscopic/robotic Fowler-Stephens with concurrent abdominoplasty.
- Genetic testing (CHRM3, HNF1B).
- Long-term follow-up of transplant outcomes.

Diagram to draw: child with wrinkled abdomen; VCUG sketch of a large bladder, urachal diverticulum, dilated triangular prostatic urethra and tortuous ureters.

Key points to remember: - Eagle-Barrett triad: deficient abdominal wall, urinary tract dilatation, bilateral intra-abdominal testes; ~95% male. - Woodard I = pulmonary hypoplasia (fatal); II = full triad with uropathy; III = mild, stable. - Most are managed conservatively; vesicostomy if drainage is needed. - Transabdominal orchiopexy by 6-12 months, often with Monfort abdominoplasty (also Randolph, Ehrlich). - CHRM3 mutation in familial cases; about one-third reach CKD; transplant outcomes are good.